LECTURE NOTES

Lecture 0: Course Introduction

AI508 - Comutervision

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Contents

1 About comutervision

Definition: Comutervision

The study and practice of learning anf computing with visual data to infer, organize, and synthesize information about the visual world.

Definition: Specific definition

The study and practice of developing and applying deep learning models for perceiving and generating visual and multi modal data.

Note that visual data refers to images, videos and other 2d/3d signals

1.1 Subfields

- Image processing (de-noising, de-blurring, superresolution, color)
- Low level vision (edge/line/corner/feature detection)
- Geometry (stereo vision, depth/pose estimation, multi-view geometries)
- **Perception** (object detection, classification, segmentation)
- **Synthesis** (image/video generation, in/out-painting, style transfer)
- **Multimodal** (contrastive language-image pre-training, vision language models)

for this course, **these** are mostly relevant.

2 Lectures

- Lecture 0 Introduction (this)
- Lecture 1 YOLO (you only look once)
- Lecture 2 Vision transformer
- Lecture 3 Diffusion
- Lecture 4 Vision language models
- Lecture 5 Segment anything